

Molecular methods for studying methanogens of the human gastrointestinal tract: current status and future directions

Prem Prashant Chaudhary¹ · Nadia Gaci¹ · Guillaume Borrel² · Paul W. O'Toole² · Jean-François Brugère¹ 

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Abstract Until recently, human gut microbiota was believed to be colonized by few methanogenic archaeal species. Much higher microbial diversity within the human gut was revealed by the use of molecular approaches as compared to routine microbiological techniques, but still, a lot remains unknown. Molecular techniques has the advantage of being rapid, reproducible, and can be highly discriminative as compared to conventional culturing methods. Some of them provide both qualitative and quantitative information. However, the choice of method should be taken with care to avoid biases. The advent of next-generation sequencing gives much deeper information from which functional and ecological hypotheses can be drawn. In this review, molecular techniques that are currently used together with their possible future developments to study gut methanogenic communities are indicated along with their limitations and difficulties that are encountered during their implementation. Moreover, the high amount of metagenomics data from the human gut microbiome indicate that this environment could be a paradigm for new directions in methanogen diversity studies and help to develop new approaches for other environments as well. Concerning humans, this should help us to better understand the possible association of methanogens with some of the diseased conditions and their peculiar distribution among age groups in human.

Keywords Methanogens · Diversity · Molecular tools · Human gastrointestinal tract · Molecular marker

Introduction

Human gut is the habitat for trillions of microbes (Eckburg et al. 2005; Qin et al. 2010; Lozupone et al. 2012) including methanogens (Miller and Wolin 1986; Liu and Whitman 2008; Sahakian et al. 2010; Gaci et al. 2014). These methanogenic archaea live in a syntrophic relationship with other microorganisms in the human gastrointestinal tract (GIT) in more than 50 % of humans (adults). Methanogens are likely absent or present at very low levels in newborns while being at a higher level in elderly people (Mihajlovski et al. 2010; Polag et al. 2014). It now becomes evident that methanogenesis is hydrogen dependent in humans and seems mainly to involve the reduction of CO₂ in most common cases of methanogenesis (Table 1). Therefore, methanogenesis might improve the energetic yield retrieval of fermentative bacteria by lowering the H₂ partial pressure in the gut together with sulfate reducing bacteria and acetogens (see for example Carbonero et al. 2012, for the consequences and bioenergetics of this hydrogenotrophy). Out of the *Methanobacteriales*, *Methanobrevibacter smithii* (*Mbs*) and *Methanosphaera stadtmanae* (*Mss*) are historically the older known human gut methanogens. Miller and co-workers reported them in the 1980s and isolated first *Mbs* (Miller et al. 1982) followed by *Mss* (Miller and Wolin 1985) from a human fecal sample. In human GIT, *Mbs* is believed to be the main archaeon in terms of prevalence and quantity; therefore, the most important reaction for hydrogen use by gut methanogens is consequently allowing to utilize 4 mol of H₂ for each mole of CO₂ converted into 1 mol of CH₄. *Mbs* can also use formate directly to generate methane. *Mss* produces methane not from CO₂ but using a different pathway, the

✉ Jean-François Brugère
jf.brugere@udamail.fr

¹ EA 4678 CIDAM, Clermont Université; Université d'Auvergne, Clermont-Ferrand, France

² Department of Microbiology and Alimentary Pharmabiotic Centre, University College Cork, Cork, Ireland

Table 1 Most frequently encountered methanogens in the human gut as known by now and their potent role in human gut physiology

Order ^a	Species	Genome	Substrate for methanogenesis	References and comments
<i>Methanobacteriales</i>	<i>Methanobrevibacter smithii</i> ^b	+	H ₂ + CO ₂ or formate Possibly CH ₃ OH	Miller et al. 1982 Samuel et al. 2007; Gaci et al. 2014
	<i>Methanospaera stadmanae</i> ^b	+	H ₂ + CH ₃ OH	Miller and Wolin 1985 Fricke et al. 2006 <i>mrtA</i> paralog instead of <i>mcrA</i> gene
<i>Methanomassiliicoccales</i>	<i>Methanomassiliicoccus luminyensis</i> ^b	+	H ₂ + CH ₃ OH/MMA/DMA/TMA ^c DMS ^d	Dridi et al. 2012 Brugère et al. 2014
	<i>Ca. Methanomethylophilus alvus</i>	+	H ₂ + CH ₃ OH ^c MMA/DMA/TMA/DMS ^d	Borrel et al. 2012
	<i>Ca. Methanomassiliicoccus intestinalis</i>	+	H ₂ + CH ₃ OH ^c MMA/DMA/TMA ^d	Borrel et al. 2013a
	Other phylotypes (Mx-2, Mx-3, Mx-4)	–	ND	Mihajlovski et al. 2010 Several phylotypes (<i>mcrA</i> marker) corresponding to different species

^a Only major species or those retrieved from several different studies are shown

^b Human isolate available

^c Experimentally proven for isolates or supposed from enrichment cultures data

^d Inferred from genomic data

MMA monomethylamine, DMA dimethylamine, TMA trimethylamine, DMS dimethylsulfide, *mcrA* gene coding the methyl-coenzyme M reductase, subunit alpha, *mtaB* gene coding the methanol:cobalamin methyltransferase, ND not determined

reduction of methanol in the presence of H₂ (Table 1, Miller and Wolin 1985; Fricke et al. 2006) and is dependent on acetate as a carbon source (Walker et al. 2006). Methanol originates from the gut bacterial degradation of pectin, mainly from fruits.

Recent advances in molecular approaches renew our vision of the methanogen diversity as well as their role in the gut: To illustrate this, genes involved in methanogenesis from methanol have been found in human gut isolates of *Mbs* (Hansen et al. 2011; Gaci et al. 2014), and therefore, methanol may also be used as a substrate for methanogenesis in the human gut for this species. Also, besides these two *Methanobacteriales*, molecular techniques coupled to phylogenetic analyses indicated that a *Thermoplasmata* lineage first identified in the rumen (Tajima et al. 2001) was also present in the human gut. It was there identified as a probable methanogenic lineage forming a new order of methanogens (Mihajlovski et al. 2008), consisting of various phylotypes, likely various species and genera or even families in the gut microbiota of elderly people (Mihajlovski et al. 2010). Members of this order were sometimes formerly grouped under rumen cluster C (RCC) (Janssen and Kirs 2008) and called *Methanoplasmatales* (Paul et al. 2012) or *Thermoplasmatales*-related methanogens. They are now officially constituting a distinguished order of methanogens called as *Methanomassiliicoccales* (Iino et al. 2013; Oren and Garrity 2013) which also possess the ability to deplete H₂ by using methanol. They seem to have a broader range of substrates for methanogenesis, for example, utilization of methylamines, as illustrated in ruminants (Poulsen et al. 2013), in termites

(Lang et al. 2015), and in humans (Brugère et al. 2014). This is very likely to be of medical importance in humans. The gut-produced trimethylamine (TMA) which originates from various nutrients (L-carnitine, phosphatidylcholine, choline,...) is considered to be the intestinal precursor of the plasmatic oxidized form TMAO. The latter is an independent risk factor of cardiovascular disease and an atherosclerosis factor (Wang et al. 2011; Koeth et al. 2013). Therefore, some strains of the *Methanomassiliicoccales* likely have a natural protective role when present in the gut and could also be supplied as probiotics (“Archaeobiotics”) in the future, to prevent atherosclerosis and also trimethylaminuria (Brugère et al. 2014).

The question is however to determine if methanogens possess other roles that are beneficial or deleterious, and answers remain still open and controversial (Gaci et al. 2014). One main reason of our present lack of knowledge is likely the absence of precise tools to study the role of methanogens in health and diseases. The different human gut methanogen species or subspecies could have different kinds of interaction within the gut microbiota and thus different relationships with health and disease (Pimentel et al. 2012; Samuel and Gordon 2006; Samuel et al. 2007; Armougom et al. 2009; Blais Lecours et al. 2014; Bang et al. 2014). Due to the stringent growth conditions of methanogens, there has been a shift toward culture-independent techniques for diversity studies. Molecular techniques are more accurate and less time-consuming which makes them now the method of choice for the investigators (Nocker et al. 2007; Brugère et al.

2009; Claesson and O'Toole 2010). It should however be kept in mind the importance of the pre-analytical steps, particularly the choice of the nucleic acid isolation method. This step can introduce a bias which is involved in every kind of molecular methods but is likely of higher importance in case of methanogens as the method can lead to variable extraction yields, not only between bacteria and methanogens, but also among methanogens themselves. The reasons are that archaea do not present usual bacterial membranes and cell wall components (Klingl 2014) and that differences among the cellular structure of gut methanogens has been observed: For example, the *Methanomassiliicoccales* are remarkably constituted of a double cell membrane as determined with electronic microscopy (Dridi et al. 2012; Lang et al. 2015). Therefore, some protocols are likely more adapted to methanogens (Dridi et al. 2009; Chaudhary et al. 2011). Once DNA is obtained, various types of molecular techniques can be applied. Figure 1 summarizes the different types of methods which are described in this review and how they can be articulated together with well-defined human populations/samples for adequate analyses. This can lead to a more accurate determination of the distribution of each of these methanogens within different populations, in order to better understand their specific role and interactions with other microbial components.

Use of classical molecular markers to decipher the diversity of methanogens

Cloning and fingerprinting techniques

Methanogen diversity in human gut has been predominantly studied by the combination of simple molecular techniques such as PCR followed by cloning and sequencing of clones as it was done earlier for other environments (Hales et al. 1996; Skillman et al. 2004). The sequences obtained are then subjected to bioinformatic analysis which results in the preparation of a phylogenetic tree that gives an estimate of the richness of methanogens in a particular sample. The most commonly used markers to study the diversity of methanogens in human gut are the 16S ribosomal RNA (rRNA) and *mcrA* genes (Scanlan et al. 2008; Mihajlovski et al. 2008; Mihajlovski et al. 2010). The *mcrA* gene encodes the α subunit of the methyl-coM reductase of the MCR-I which catalyzes the last step of methanogenesis (methane release from coM) (Fig. 2, see Thauer 1998; Deppenmeier et al. 1999; Thauer et al. 2008 for description of the methanogenesis biochemical pathways and related energy conservation). However, the human gut inhabitant *Mss* does not possess a MCR-I complex, but only the type II (MCR-II), which is composed of subunits encoded by the *mrt* genes (Fricke et al. 2006). Therefore, caution should be taken to detect this

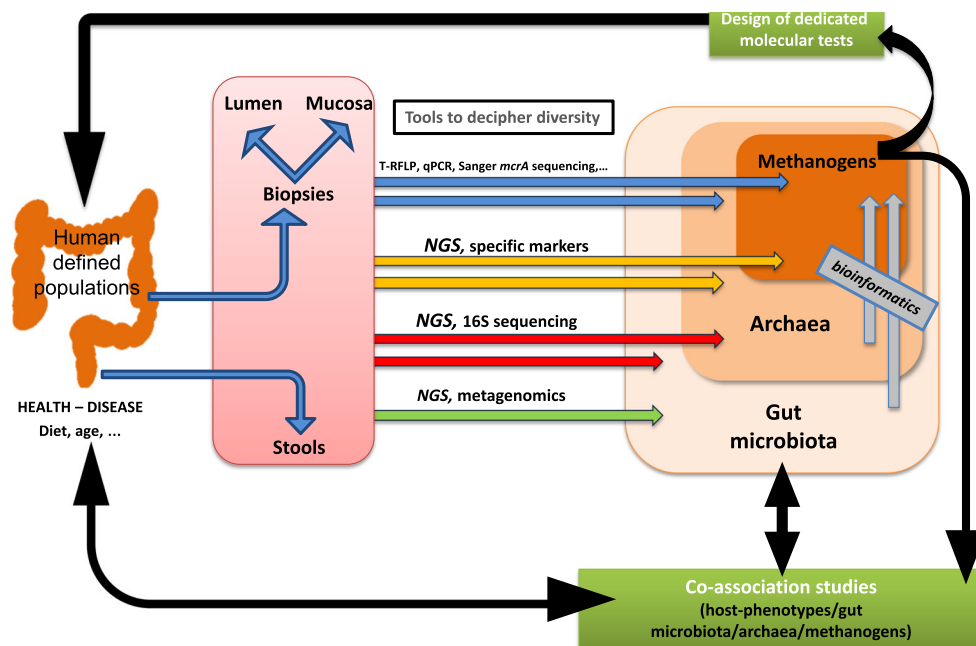
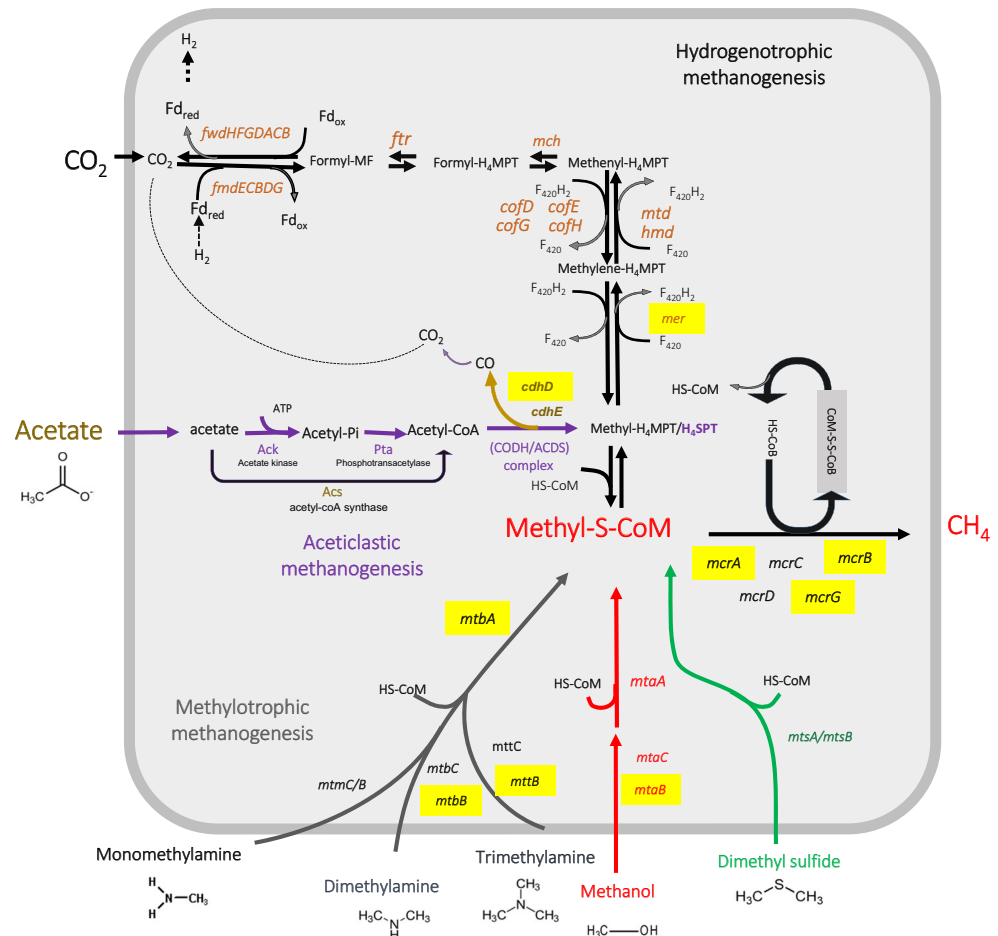


Fig. 1 Different scales to address the gut methanogen diversity. Two types of samples are generally collected for the diversity and association studies from humans, i.e., biopsies which can further be subdivided based on the region of collection (for example, lumen, mucosa, and stool samples, as shown in the pink box). Diagrammatic representation of the traditional and future molecular tools used in order

to study archaeal or methanogenic diversity with their degree of penetration is shown in this figure. Data obtained from these studies can be used next for co-association studies between methanogens and defined population and to elaborate specific diagnostic molecular tools, like PCR/qPCR tests to address specific questions

Fig. 2 Molecular markers to study gut methanogens. All the three possible methanogenic pathways are represented in this figure (adapted from Thauer 1998; Deppenmeier et al. 1999). Although, there is no evidence of aceticlastic methanogenesis in human gut. Genes which have been shown to be possible molecular markers for the hydrogenotrophic and methylotrophic methanogens in order to study their diversity are highlighted by yellow color. Several subunits of *mcr* operon encoding the methyl-coM reductase of type I (present in quite all the methanogens, but absent in *M. stadtmanae*) can be used to address the methanogen diversity, with *mcrA* which has been used as a universal molecular marker of methanogenesis



archaeon based on this marker. These methods are much time-consuming, laborious, and costly as large numbers of clones need to be screened in order to cover the maximum diversity, together with the sequencing technology (typically the Sanger method). This can be reduced if combined with an oriented cloning and/or a fingerprinting technique (Mihajlovski et al. 2008), which leads to an easiest selection of clones (based on restriction fragment length polymorphism (RFLP) for example).

To our knowledge, there are few reports of the application of denaturing gradient gel electrophoresis (DGGE)/temporal temperature gel electrophoresis (TTGE) (Muyzer 1999) in studying human gut archaea or methanogens specifically (Abell et al. 2006; Nam et al. 2008), despite having already been extensively used for other environments (see for example Nakamura et al. 2009; Chaudhary et al. 2014). One major limitation of this technique relies on its inability to differentiate minor base pair differences in two sequences and thus may result in single band for different microbial strains. Another possible technique for studying gut methanogens is denaturing high-performance liquid chromatography (dHPLC) which allows a greater number of samples to be processed in a more convenient way. It is typically used to detect mutations by

their property to form heteroduplex between wild-type and mutated DNA strands. Heteroduplex molecules are separated from homoduplex molecules by ion-pair, reverse-phase liquid chromatography on a special column matrix with partial heat denaturation of the DNA strands (O'Donovan et al. 1998; Oefner and Underhill 1998). This approach can also be used in microbial ecology (Barlaan et al. 2005) encompassing the gut environment (Goldenberg et al. 2007). Here, PCR amplification (usually targeting 16S genes) is performed similarly to DGGE, that is with a GC clamp to promote separation among different amplicons (Illmer et al. 2014). Till now, there has been no report of the use of this technique in studying gut methanogen diversity. This might be due to the high cost of the equipment and columns or high degree of optimization required for each reaction to achieve the highest sensitivity of the results.

Terminal restriction fragment length polymorphism (T-RFLP) technique involves the amplification of usually the 16S rRNA gene or any target gene, *mcrA* gene in case of methanogens. Out of the primer pair used, one primer is fluorescently labeled with some dye. These amplified products are then subjected to restriction digestion with one or several restriction enzymes. An automated sequencer generates an

electrophoregram based on the sizes of the T-RFs in a given sample. Vianna and co-workers used T-RFLP to study the prevalence and distribution of methanogens and possible associations with bacteria in oral biofilms, and they found the dominance of *Methanobrevibacter oralis* along with one unique phylotype (Vianna et al. 2009). However still, this technique has been underestimated in studying the methanogen diversity in human gut. One of the limitations is the impossibility to distinguish two sequences when they share a common terminal restriction site, as it will result in the same peak during the capillary electrophoresis and thus leads to ambiguous elucidation of microbial diversity. The use of several restriction enzymes in parallel reduces this risk. Another problem is the lack of well-defined databases to compare and assign the obtained T-RFs to a species. In spite of this, T-RFLP is still considered a reliable and high-throughput technique to monitor the diversity of microbial populations in different environmental samples (Scavino et al. 2013; Conrad et al. 2014; Mach et al. 2015).

Suppressive subtractive hybridization (SSH) allows the identification of one bacterial genome from another closely related one on the basis of the identification of the DNA fragments that are uniquely present in only one genome (Peng et al. 2011). A comparison is made in the genomic content of the closely related species by using a strain-specific genomic DNA, next cross-hybridized and then compared with known gene or protein sequence data. Finally, it allows establishing a functional relationship between these DNA fragments and the compared genomes/DNA content. This technique can also be used as a novel method to identify the unknown methanogen diversity within the human gut, at least for determining the gene repertoire that differs among strains (see below).

Detection and quantification of methanogens by molecular techniques

Quantitative real-time PCR (qPCR) is one of the most sensitive, reliable, and reproducible techniques in order to determine any particular microorganism in an environment. The principle of this technique is based on the real-time detection of fluorescence of the reporter molecule during the course of the specific amplification by PCR. The threshold value of the fluorescence depends upon the amount of gene copy number in the sample, i.e., the higher the gene copy number in the starting sample, less will be the threshold value and vice versa. This technique has been widely used to study the diversity of methanogens and other microorganisms in ruminants and other environmental samples (see for example Ouwervkerk et al. 2002; Chaudhary et al. 2012; Zeleke et al. 2013), but there are only few reports that deal with the quantification of archaeal or methanogenic population in human gut (for example, see Abell et al. 2006; Dridi et al. 2009; Million et al. 2012; Kim et al. 2012). Zhang

and coworkers used qPCR technique to study and compare the shifts in the gut microbiota in obese and after gastric bypass patients as compared to the control ones (Zhang et al. 2009). It was also used to determine the abundance and richness of mucosa-associated hydrogenotrophic microbes in healthy human colon (Nava et al. 2012a), which showed that all the three hydrogenotrophic groups of microbes including methanogens, i.e., *Methanobrevibacter* spp., *Mbs*, and members of the order *Methanomicrobiales* were present in human colonic mucosa (Nava et al. 2012b). Both relative and absolute quantifications are possible by the use of this technique. For an absolute quantification, standards need to be prepared from the known concentration of plasmid samples. Its high cost and difficulty to determine two or more targets within one assay is however limiting. Determining a higher number of targets in one assay can be achieved using microbial ecological microarrays, but the quantification remains relative. Several DNA chips have been developed specifically toward the human gut microbiota based on 16S targets: This encompasses phylogenetic DNA chips, like the human intestinal tract chip (HITChip, Rajilić-Stojanović et al. 2009) which is likely the most used, but does not encompass the detection of archaeal species of the gut. This is also the case of the human gut chip (HuGChip, Tottey et al. 2013), despite being designed with explorative probes (can detect unknown DNA sequences and affiliate them at the family level). Inversely, DNA chips specifically designed toward methanogens could help for this purpose, like the ANAE ROCHIP (Franke-Whittle et al. 2009). The fact that it is designed toward already known methanogens can be however an important limitation, and this kind of microarray had never been used for investigation of the human gut methanogenic community. At last, another DNA microarray solution resides in functional profiling of human gut microbiota with DNA array like the human microbiome chip (HuMiChip, Tu et al. 2014) as it possesses probes toward methane metabolism.

Fluorescence in situ hybridization (FISH) is a quantitative technique that can be used in microbial ecology for the specific detection/visualization of microorganisms. It is based on fluorescent nucleic acid probes which bind to the complementary regions on the microbial chromosome, finally analyzed by fluorescence microscopy. Numbers of probes are available for the quantification of methanogens in ruminants and environmental samples, but this technique is still not much used in the quantification of methanogens in human gut. We came across only very few studies in humans where FISH has been used for the quantification of methanogenic archaea. One of this kind of study was conducted by Lepp and his coworkers where they used FISH for the detection of archaeal cells at subgingival sites in periodontitis patients (Lepp et al. 2004). However, for the implementation of this technique in studying gut methanogens, a detailed knowledge about the gut methanogenic community should be there so that more specific probes could be designed for appropriate studies in this kind of

environment. There are also some limitations when FISH is implemented on archaea, like the low copy numbers of 16S rRNA genes found per genome and their different cell wall and membrane constitution compared to bacteria. So, more sensitive methods like catalyzed reporter deposition (CARD)-FISH should be used in order to detect gut methanogens more precisely. In 2008, Kubota et al. tested several permeabilization protocols using different enzymes and comes out with the conclusion that PeiW treatment is the most effective one for CARD-FISH application to methanogens (Kubota et al. 2008). Independently, methanogens were believed till recently to possess one cofactor supposed to be necessary for all methanogens, F_{420} . Despite that its presence in few other microbes has already been reported, but at lower levels, an excitation at 420 nm and emitted fluorescence permit the detection and counting of methanogens (Doddema and Vogels 1978). However, this F_{420} factor is very likely absent in members of *Methanomassiliicoccales* order (Borrel et al. 2014a).

In-depth sequencing of molecular markers

Massive parallel sequencing, further described as next-generation sequencing (NGS) methods, provides a mean to rapidly and deeply sequence DNA from a sample. Various NGS methods have been used (454 pyrosequencing, Illumina technology), and some new ones are also available now (i.e., PacBio and IonTorrent), each with their own specifications and limitations (Roh et al. 2010; Weinstock 2012). Due to the high amount of data generated in a single run of sequencing, these technical approaches are useful to access information about diversity at various scales, from genomic data of methanogens to metagenomics data (see next chapter). It can also be performed on specific/dedicated molecular markers of methanogens/methanogenesis (Fig. 1, yellow, red, and green arrows). However, several factors should be kept in mind when designing a microbiome study (Goodrich et al. 2014a).

The use of 16S rRNA gene as a biodiversity marker for methanogen studies in samples relies mainly if not exclusively on universal primers that have been designed through comparative 16S sequences of bacteria and can therefore introduce a bias toward methanogen detection, by being less or not efficient at all (as also mentioned above for probes used with phylogenetic microarrays). Recently, a prokaryotic universal primer pair was proposed for a simultaneous analysis of bacteria and archaea (Takahashi et al. 2014). It was designed by targeting V3–V4 16S region on the Illumina MiSeq sequencer and allows the detection of more than 94 % of archaeal and 98 % of the bacterial rRNA genes theoretically, according to the Ribosomal Database Project (RDP) database. It was used in studying pig gut microbiota, and bacterial and archaeal components were detected in one sequence reaction: For these last ones, *Methanobacteriales*, but also *Thermoplasmata*

(very likely *Methanomassiliicoccales*, see below) were determined. Alternatively, a strategy based on several couples of primers, allowing a better coverage of each of the three domains of life (*Eukarya*, *Bacteria*, and *Archaea*) could allow to limit bias in the detection of every component of a microbiota (encompassing methanogenic archaea). This has been recently developed and used for the determination of rumen microbial communities with 16S amplicons generated by several primers next sequenced together (Kittelmann et al. 2013). Such an approach would be very likely interesting concerning studies on the human gut microbiome as the cooccurrence of every microbes of the ecosystem can be addressed in a simple and rapid way.

When addressing the question of the methanogen diversity itself, it would be of great interest to limit generated data to only methanogens. Unfortunately, methanogenesis seems to have been lost several times in the evolution of the *Euryarchaeota* (Baptiste et al. 2005; Borrel et al. 2013a; Raymann et al. 2015) leading to sparse methanogenic lineages separated by non-methanogenic orders. Among these lasts, some of these archaea may be found in the gut microbiota, especially *Halobacteria* (Nam et al. 2008; Oxley et al. 2010). Consequently, non-methanogenic 16S sequences could be detected, a problem that can be overcome by fine annotation/phylogenetic studies. Inversely, the use of 16S primers with a narrow coverage could limit the real methanogenic diversity, for example, of *Methanomassiliicoccales* located within the class *Thermoplasmata*. In the seminal work of Eckburg and his team based on cloning and sequencing by the Sanger method, two types of primers were used: one for the bacterial component and another for the archaeal part (Eckburg et al. 2005). This approach was performed on different types of samples from the GIT, encompassing mucosal tissue collected from three individuals. Only one operational taxonomic unit (OTU) was detected for archaeal members among the 1524 obtained sequences, corresponding to *Mbs*. This strongly, but wrongly conveys an idea of low methanogenic diversity within the human gut. Increase in the availability of sequencing data for the third domain, i.e., archaea, leads now to specific primers which are able to detect a broader part of archaeal population. This approach has been successfully used recently for human gut archaea (Hoffmann et al. 2013). In this study (dedicated to fungi and archaea of the human gut), various and not yet recovered gut archaea were detected, larger than only methanogens/*Euryarchaeota*. This encompasses *Thaumarchaeota* of the *Nitrososphaera* genus. Also, this study revealed euryarchaeotes affiliated to two different genera of the *Thermoplasmata*: the *Thermoplasmatales* *Thermoplasma* spp. and the *Thermogymnomonas* spp., thermoacidophilic and aerobic archaea. Surprisingly, no *Methanomassiliicoccales* were retrieved from this study. As mentioned above, the *Methanomassiliicoccales* belongs to the class *Thermoplasmata* with other non-methanogenic order

(for example, *Thermoplasmatales*) and are therefore phylogenetically related with each other. Together with the fact that no other studies indicated the presence of unambiguously determined *Thermoplasmatales* in the human gut, it is likely that a bias in the affiliation had been made. In fact, in this rapidly moving area of methanogens, the databases used for the analysis should be regularly updated and encompass sequences of non-cultured/isolated archaea and for which a fine and strong phylogenetic affiliation has been obtained. As an example, in March 2015, the RDP Classifier (RDP release 11, update 3) (Wang et al. 2007) did not identify the *Methanomassiliicoccales*, even if several genomes had already been published and were deposited in GenBank (Gorlas et al. 2012; Borrel et al. 2012; Borrel et al. 2013b; Lang et al. 2015). At the time we submitted this article, it still affiliates *Methanomethylophilus* spp. and *Methanomassiliicoccus* spp. to *Thermoplasma/Thermogymnomonas* spp., but this should be improved in the coming releases of the RDP Classifier. Alternatively, the use of fine-annotated and dedicated 16S database should provide a way to circumvent this kind of problems. This has recently been undertaken for methanogens of the GIT of ruminants (Ruminal and Intestinal DB (RIM-DB), Seedorf et al. 2014). This DB considers also other intestinal environments such as the human and termite intestinal tract, in which methanogens are particularly of interest in the global warming context (Schmitt-Wagner and Schmitt-Wagner and Brune 1999; Brune 2014). RIM-DB contains more than 2300 chimera-checked 16S sequences and allows a species-level taxonomic assignment. *Methanomassiliicoccales* are represented by 195 sequences distributed in 17 groups, possibly different genera, which highlights the diversity of *Methanomassiliicoccales*. This tool can be implemented in QIIME (Caporaso et al. 2010) as a reference database when affiliating sequences from amplicon-based NGS of 16S rRNA gene.

Considering all of this, the sequences supposed to come from *Thermoplasmatales* in the study of Hoffmann and co-workers (Hoffmann et al. 2013) and also in various other studies from other digestive environments are probably methanogens. Another way to limit this possible artifact would be the use of *mcrA* gene marker, specific to methanogens (i.e., absent in *Thermoplasmatales*) and for which phylogenetic affiliation is possible (Luton et al. 2002). Other alternative molecular markers are also possible, as well for methanogens than for (Eury)archaea.

Marker-based alternatives to study methanogen diversity

Some molecular markers have already been used for the detection of specific methanogens in human gut, for example, the 16S rRNA and *rpoB* genes (Dridi et al. 2009). However, there still feels the need of alternative molecular markers in

order to target specific functions of the methanogens or to discriminate different subspecies.

The genomic sequence neighboring the 16S and 23S gene sequences (internal transcribed spacer (ITS)) is less subjected to high selective pressure compared to these two genes, which confers to it a high polymorphism in sequence and length. It can therefore be very useful for determining the diversity of methanogens using automated ribosomal intergenic spacer analysis (ARISA) (Fisher and Triplett 1999; Ciesielski et al. 2013). However, there is no unique transcription unit for 16S and 23S rRNAs in *Methanomassiliicoccales*: These two genes do not form an operon as usually observed in bacteria and other methanogens and are scattered in the genome (Borrel et al. 2014b). Consequently, there is no ITS/a lack of a small DNA region amplifiable by 16S/23S primers. Therefore, the use of ARISA would lead to a strong bias toward the detection of one of the major methanogenic populations in human gut.

In addition to *mcrA* and 16S rRNA genes, chaperonin type II markers (also called as thermosome) were also used as a target to study the phylogenetic diversity of archaea. Type II chaperonin sequences are considered to be more diverse than the previously used markers and thus might serve as a better marker for the diversity studies (Chaban and Hill 2012). Phylogenetic trees constructed by using chaperonin type II markers are in well agreement with 16S rRNA-based trees, but rather, they provide a more detailed resolution of the closely related groups, or it could help to discriminate different subspecies. When tested along with *mcrA* and 16S rRNA genes to study the archaeal diversity in a cattle rumen, *Mbs*, *M. ruminantium* and *Mss* were successfully detected as the dominant members of the methanogen community with all the three markers. This marker-based diversity analysis still needs to be used for methanogenic communities inhabiting the human gut. Several other molecular markers have been studied and validated recently for the diversity analysis of methanogens (Romaniuk et al. 2014). These markers and the place of enzymes they are coding for in the methanogenesis pathways are indicated in Fig. 2. This includes *mcrB* and *mcrG* which are theoretically necessary to all the methanogens as they encode subunits of β and γ subunits of the methyl-coenzyme M reductase MCR-I. However, as mentioned above, MCR-I is absent in one human gut methanogen, *Mss*, where it is replaced by the sole presence of a type II methyl-coenzyme M reductase (Fricke et al. 2006).

Other tested molecular markers are restricted to a kind of the three methanogenic pathways, i.e., the methylotrophic, acetoclastic, and hydrogenotrophic ones (Fig. 2). For the methylotrophic methanogenesis, some markers were used successfully (Borrel et al. 2013a; Romaniuk et al. 2014; Blais Lecours et al. 2014) like *mtaB/mtaB1* (methanol-5-hydroxybenzimidazolylcobamide Co-methyltransferase) for methanogenesis from methanol and *mtbA* (methylated

[methylamine-specific corrinoid protein]: coenzyme M methyltransferase) or *mtbB1* (dimethylamine-corrinoid protein Co-ethyltransferase) for methanogenesis from dimethylamine (DMA). For the acetoclastic pathway (mentioning that no methanogen exhibiting acetoclastic mode of methanogenesis has been detected to date in the human gut), *cdhD1* coding the δ subunit of acetyl-CoA decarboxylase/synthase complex is a possible molecular marker (Romaniuk et al. 2014). At last, for the hydrogenotrophic methanogenesis, various markers are possible, and among them, some have been successfully used like the *mer* genetic marker encoding the F₄₂₀-dependent methylene-H₄ MPT reductase (Romaniuk et al. 2014).

However, caution should be applied when using these kinds of molecular markers: They can have a broader distribution than methanogens only or inversely, be absent in some gut methanogens genomes, due to the diversity of methanogenesis encountered in the human gut. For example, it has been shown that the *Methanomassiliicoccales* are not able to complete methanogenesis from CO₂ and H₂ and lack all the specific genes encoding necessary enzymes of this complete metabolic pathway (i.e., *fwdHFGDACB*, *fmdECBDG*, *fr*, *mch*, *cofDEGH*, *mtd*, *hmd*, *mtrADCBAEFGH*, and *mer*) (Borrel et al. 2013a). Therefore, the use of *mer* is likely insufficient for gut methanogens. It has however to be noticed that the use of the *mtaB* gene (methanogenesis from methanol) has been successfully employed for a description of the whole *Methanomassiliicoccales* order (Borrel et al. 2013b). This gene is present in *Mss* (Fricke et al. 2006) and seems to be also present in *Mbs* strains from the human gut (Gaci et al. 2014) even if they have not been described as methanol-using methanogens: Therefore, *mtaB* gene may be very useful as a marker gene to decipher gut methanogen diversity but is inversely a possible way to introduce a bias in studies if used alone. In conclusion, these genes are strong molecular markers to study the diversity of methanogens in the human gut, but one should be used in conjunction with another to cover the broad range of methanogenesis pathways.

Future directions to shift from description of diversity onto function of methanogens

Coupling molecular tools with culturing techniques

All the techniques described above allow diversity studies of the methanogens and their distribution/changing during lifetime. However, it remains very hard to decipher functions other than methanogenesis itself. Interventional studies on the natural habitat (the human gut) are ethically and technically problematic. Altogether, this argues for the use of ex vivo/in vitro studies, i.e., culture-based approaches on isolated microbes or complex communities in in vitro systems simulating the gut environment, which can be further associated with molecular tools.

Recently, there has been a renewed interest in culturing methods for the detection of species (Goodman et al. 2011; Ma et al. 2014; Rettedal et al. 2014). Lagier et al. introduced the concept of culturomics by combining the mass spectrometry approach with existing sequencing (Lagier et al. 2012). They applied this technique on the stool samples of two lean Africans and one obese European in order to study/compare the gut microbial populations simultaneously along with traditional metagenomic analysis. Culturomics identified 340 bacterial species, and the routine metagenome analysis of the same samples yielded 698 phylotypes: This included 282 known species, 51 of which overlapped with the microbiome obtained from culturomics (Lagier et al. 2012). Thus, culturomics can provide complementary details on the gut microbiota. This is a much open approach that detects viable bacteria and that is theoretically not restricted to bacteria only. However, no archaeal species were retrieved from this study while five fungi were detected. This approach stills needs to be applied in order to study the diversity of methanogens specifically in the human gut. Advantages includes the identification of species which are present in low number in the ecosystem, i.e., minority populations, which is of great interest for gut methanogens, but is so far strongly limited by difficulties in culturing methanogens.

Several progresses have been made in simulating the gut microbiota environment in order to follow the metabolic/cellular interactions within the gut microbes. To date, various in vitro systems have been developed (Venema and van den Abbeele 2013), the first ones more than 25 years ago (Macfarlane et al. 1989; Macfarlane et al. 1998). Interestingly, methane production was observed in the three-stage in vitro systems developed by MacFarlane and Gibson after an inoculation with *Mbs*-positive stools, indicating that methanogens could survive and be metabolically active in this in vitro model (Gibson et al. 1988). However, to our knowledge, all the systems developed so far uses an imposed anaerobic atmosphere to microbes (commonly, a continuous flush of CO₂/H₂ or N₂/CO₂ used as oxygen scrubber), a possible bias toward microbe interactions and metabolisms: In fact, other microbial populations may be sensitive to gas level produced by the ecosystem like H₂S/HS⁻. This is of importance considering the probable competition between hydrogenotroph populations, particularly SRBs and methanogens. To overcome this problem, an in vitro continuous fermentation concept named Environmental Control System for Intestinal Microbiota (ECSIM) has been developed which allows culturing microbiota under the atmosphere developed by the microbiota itself (Brugère et al. 2011). This has been first declined as three parallel fermentation systems, allowing concomitant comparative studies (Feria-Gervasio et al. 2011). Recently, a derived model of the one developed by MacFarlane et al. (1998) was obtained, forming a three-stage continuous system simulating three gut regions from one common inoculum (3S-

ECSIM) (Feria-Gervasio et al. 2014). An exhaustive comparison of the microbial components of the initial stool inoculum revealed a high correlation at the taxonomic level of bacterial families using the HuGChip validating this approach (Feria-Gervasio et al. 2014). Importantly, the cultivation of a methanogenic stool inoculum has been recently achieved in the ECSIM system, encompassing methane production and *Mbs* yields similar to what was observed in the original sample (Tottey et al. 2015). This has also been coupled to specific inhibition of methanogens and to the in vitro conversion of a non-methanogenic stool sample into a *Mbs*-positive one (Tottey et al. 2015), permitting to achieve what failed in animal models (Hansen et al. 2011; Goodrich et al. 2014b).

Therefore, the next step will be to extend this model for studying gut methanogens to various gut microbiota and various methanogens. From this, microbial interactions involving gut methanogens could be deciphered. This could be helped, for example, with stable isotope probing (SIP), for which in vitro systems fit technically well. The SIP method is now widely used in microbial ecological studies for tracing fluxes of nutrients in biogeochemical cycling by microorganisms. This technique can further be used to check the predominant members of the mixed community which are metabolically active in utilizing a particular substrate for their growth (Radajewski et al. 2000; Dumont and Murrell 2005). The labeled nucleic acids are separated from the unlabeled ones by density gradient centrifugation methods. This labeled nucleic acid pool can further be analyzed using NGS, allowing population studies based on the 16S marker.

Sequence enrichment by solution hybrid selection capture method

Hybridization capture technologies, and among them those performed in solution phase, allow a targeted exploration of metagenomes. The solution hybrid selection (SHS) capture method has been proposed (Gnirke et al. 2009; Tewhey et al. 2009) to enrich and select known parts of the human genome for resequencing. Adaptation of the gene capture for microbial ecology allowed a better taxonomic and functional description of the studied microbial communities including the rare biosphere and unknown sequences (Denonfoux et al. 2013). It involves the enrichment of the target gene fragments out of a complex environmental genomic mixture, without PCR biases and direct metagenome sequencing limitations. The success of a gene capture experiment in environmental microbiology strongly depends on the high-quality probe set encompassing variant specific and explorative probes to reveal novelties (Parisot et al. 2014) coupled to adapted databases (Jaziri et al. 2014). Additionally, this innovative strategy for microbial ecology allows recovering of large DNA fragments, therefore extending beyond the initial targeted biomarker gene sequence and thus facilitating the discovery of new genes or

genomic organizations. Denonfoux and coworkers developed this approach to detect methanogens in deep water layer of a meromictic lake (Denonfoux et al. 2013), by targeting *mcrA* sequences with explorative probes (Dugat-Bony et al. 2012). The enrichment performance (around 50 %) demonstrated the relevance of such method compared to a random shotgun metagenomic approach (0.003 %). Qualitatively, this strategy outperforms the barcoding approach allowing the detection of 17 OTUs not recovered by the amplicon strategy, and among them one *Methanomassiliicoccace*, while 24 OTUs were shared. Gene capture by hybridization gave better results when compared with amplicon sequencing and direct metagenome analysis methods, resulting in the detection of much broader gene diversity. Full-length targeted markers and adjacent regions could be rapidly reconstructed giving new insights on methanogens present in environmental samples and associated metabolic functions. More recently, this gene capture strategy has allowed the uncovering of complete cDNAs to access directly to activities (Bragalini et al. 2014). Therefore, these targeted sequencing methods could be useful in the future to study the gut methanogens in humans, enabling in-depth analysis for each sample, and high throughput, when associated with NGS.

NGS for genomics of isolated strains of methanogens

Complete genome data is of course the gold standard for comparison of strains/species of methanogens and allows to reconstruct metabolic pathways and to hypothesize features/properties/functions. Recently in 2011, pan-genome analysis of dominant gut methanogen *Mbs* has been performed in monozygotic and dizygotic twins, and the authors found out that 987 genes were conserved in all strains and 1860 genes were variably represented (Hansen et al. 2011). Pan-genome of *Mbs* also consisted of 101 adhesin-like proteins (ALPs) which showed strain-specific differences in expression and responsiveness to formate. So, it can be postulated that *Mbs* strains use their different repertoires of ALPs to create diversity in their metabolic niches, by allowing them to establish syntrophic relationships with bacterial partners with differing metabolic capabilities (Hansen et al. 2011). This study highlights the need of highly discriminating tools as even a same species (here *Mbs*) can show a high genetic variation among strains retrieved from the same environment (the human gut), so may have variable influence on the host physiology/health. As it relies on a time-consuming first step for strain isolation and that some gut strains of the *Methanomassiliicoccales* order still resist to isolation (our and other works), this direct genome sequencing and genome comparison is however still limited. Single-cell genomics of gut microbes, and more specifically of archaea, will be very likely the best method in the future (Blainey and Quake 2014), but before this time has

came, other approaches are used concomitantly with the exploration of other directions able to generate genomic data.

NGS for genomics of enrichment cultures

When not directly accessible (means isolated), enrichment cultures may be first performed to increase the relative amount of the methanogens by specific conditions (antibiotics, methanogenesis substrates, gas,...). Using methanogen-positive stools and several rounds of culture with H₂ and methanol, metagenomics by shotgun sequencing was performed directly on these consortia and led to the genome characterization of two methanogens present in the human gut, the *Methanomassiliicoccales* “*Candidatus* Methanomethylophilus alvus” Mx1201 and “*Candidatus* Methanomassiliicoccus intestinalis” Issoire-Mx1 (Borrel et al. 2012; Borrel et al. 2013a). Moreover, the mean coverage (respectively of 36.9× and 42.7× for “*Candidatus* Methanomethylophilus alvus” Mx1201 and “*Candidatus* Methanomassiliicoccus intestinalis” Issoire-Mx1 from a quarter plate of a 454 GS FLX Titanium run) was sufficient to obtain the complete draft genome sequence of these two organisms. This strategy was also recently used for a fourth *Methanomassiliicoccales* genome, from a methanogen of the termite gut (*Candidatus* Methanoplasma termitum, Lang et al. 2015). Data generated by this approach allowed the first whole-genome comparisons among three human strains of the *Methanomassiliicoccales* (Borrel et al. 2014b) from which some important genomic, phylogenetic, and biochemical features of this so far unknown methanogenic lineage were deciphered. Of note, these works provided important clues about the use of methylamines in the human gut by these strains, due to (i) a particular genetic machinery for the translational addition of the 22nd amino acid pyrrolysine (Pyl) (Borrel et al. 2014a) and (ii) the presence of genes encoding Pyl-containing methyltransferases, which participate in the methylotrophic methanogenesis from methylamines (see Krzycki 2004; Ambrogelly et al. 2007 and Gaston et al. 2011 for a description of this expansion of the genetic code particularly in some methanogens).

Genome reconstruction of methanogens from metagenomics studies

NGS technology allows an in-depth sequencing of a sample. Therefore, it may be possible to extend the method described above with enrichment cultures on DNA retrieved from fecal matter or gut biopsies. Theoretically, obtained metagenomics data from one sample contain enough sequence data to allow the alignment in contigs/genomes even at low proportion of some microorganisms. Due to the various level of each microbe/genome within an environment, there is no uniform coverage of all sequence data, and this property

may be used to filter sequences based on sequence frequencies to facilitate the assembly of sequences. This assumption should be modulated as there is no uniform coverage in sequencing depth across a genome in practice. Other difficulties rely on similar repeat regions present in various genomes of one environment and also to sequencing errors that becomes more difficult to handle with when associated to a low coverage of each genome.

This approach was recently shown successful for a new candidate phylum basal to *Cyanobacteria*, whose members have been found 10 years ago in the human gut (Ley et al. 2005). Using gut metagenomics data from three samples composed each of around 115,000,000 reads, the authors were able to obtain four complete genomes and three partial genomes, one of which being near to complete (Di Rienzi et al. 2013). In absence of cultured representatives, it was so far impossible to speculate about the role and properties of these organisms that are prevalent in herbivorous mammals and humans with a vegetarian diet, showing therefore a similar technical problem than the one encountered with *Methanomassiliicoccales*. Using this powerful approach and data inferred from the reconstructed genomes, it was deduced, for example, that this clade (candidate phylum “*Ca. Melainabacteria*”) produces H₂ through a fermentative metabolism, has the capacity to fix nitrogen, and synthesizes several (B and K) vitamins. It however needs to be mentioned that *Melainabacteria* were estimated to form a large part of the community, estimated to about 2 to 4 % which has likely greatly facilitated this approach. Considering this, and due to the lower relative abundance of human gut methanogens, this approach would necessitate a greater number of reads per sample. Alternatively, if the coverage is too low, it is conceivable to pool the metagenomics data from several samples, assuming that the considered methanogen may be shared by several individuals; this may lead to potent mosaic genomes, referred as metagenomic species (MGS) or metagenomic operational taxonomic units (mOTUs) (Sunagawa et al. 2013; Nielsen et al. 2014). Among the 7381 MGS derived from 396 human stool samples of the MetaHit project, one is attributed to a methanogen species: MGS1088 was retrieved from the distal human gut microbiota of the sample MH0012 by an abundance covariance strategy. Sequencing data are composed of 67 contigs varying from 703 to 151,917 bp (median size=14,427 bp) and further organized in 32 scaffolds (genbank NZ_CBKQ010000000). It represents a total of 1,631,477 bp, from which 1625 coding DNA sequences (CDS) were deduced. It was annotated as a *Methanoculleus* sp., from the *Methanomicrobiales* order, but when checked in detail, these sequences correspond to a species of the *Methanomassiliicoccales* order, more precisely a sister species/a different strain of “*Ca. Methanomethylophilus alvus* Mx1201” (Borrel et al. 2012) with 99 % similarity on the

whole genome (unpublished result). These results highlight the power of such an approach.

The assembly of sequences from metagenomic studies may also be helped by existing genomic data of several related subspecies/species that can be used as a backbone to recover contigs from metagenomes that are consequently related to the reference genome(s). This bottom-up genomic reconstruction may also be conducted with specific markers of the methanogens (*mcrA*, *mtaB*, etc., see above) to recover contigs that are specific to methanogens. As an example, we are currently retrieving genomic data from contigs of human gut metagenomes that harbor some genes of methanogens, and among them, those more specifically belonging to the *Methanomassiliicoccales*. This leads to new partial and total genomic data and allows detection of new variants/subspecies/species of this very poorly known lineage (unpublished data).

Concluding remarks

Detailed knowledge of gut methanogens involves the more in-depth study of the methanogenic diversity, their symbiotic relationships with other microbial groups, and various other factors that could affect the microbial communities (host genetics, diet, age,...). Molecular approaches based on DNA and RNA examinations are among the most reliable, sensitive, and less time-consuming techniques to determine their diversity. In order to study uncultured methanogens in human gut, future knowledge will likely come from bioinformatic studies of metagenomics data, whose interpretation will need to be confirmed by dedicated, strain/species-specific molecular tools on relevant population samples (Fig. 1). Altogether, these techniques will provide a gateway for gut microbiologists for designing effective and novel strategies which will help them in understanding the role of these methanogens in human health. In our opinion, traditional culture-based techniques are mandatory to be used in conjugation with these advance molecular/bioinformatic techniques to address some peculiar and important points like their interactions with the immune system or the host physiology.

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